SML Sampling

User manual for the scripts

SML_sampling_example_with_6_Italy_soil_units.R

Kriging_example_with_6_Italy_soil_units.R

Optimal selection of soil monitoring sites (Soil Monitoring Level) on a 1 km grid: customisation and execution guide

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1. Purpose and overview

The workflow selects the sampling sites of a soil monitoring network for a test area made of 6 domains (soil unit x land use) grouped in 2 zones: Toscana meridionale (60.4_Cropland, 60.4_Forest, 60.7_Cropland) and Emilia orientale - delta del Po (BE_Cropland, BA_Cropland, DO_Cropland).

The sample is optimised with the SamplingStrata package (genetic algorithm + Bethel allocation) so that the two target variables, soil pH (PH) and soil organic carbon (SOC), are estimated with a coefficient of variation (CV) of at most 5 % in every domain. Strata are built on two auxiliary variables: clay content (ARG) and elevation (DEM). Existing LUCAS points can be forced into the sample.

The workflow is split into two **independent** scripts:

Script	Role	When to run it
SML_sampling_example_with_6_Italy_soil_units.R	Sampling design: frame, LUCAS points, stratification, allocation, sample selection, outputs.	Always.
Kriging_example_with_6_Italy_soil_units.R	Spatial parameters: variogram fitting on point survey data; writes legend.csv (fitting and range per soil unit), the variogram plots and a reliability report.	Only to (re)estimate the spatial parameters, before a sampling run with spatial_component = TRUE.

The two scripts communicate only through the file **legend.csv**: the kriging script writes it, the sampling script reads it. The sampling script never runs the kriging script, and legend.csv can also be written or edited by hand.

The flag `spatial_component` of the sampling script chooses the design:

spatial_component	Phase 3 (parameters)	Phase 5 (initial solution)	Phase 6 (GA, optimStrata)	Phase 6b
TRUE	fitting and range of each soil unit read from legend.csv	KmeansSolutionSpatial with the soil-unit fitting/range	soil-unit fitting/range	performed
FALSE (default)	fitting = 1, range = 0 everywhere; legend.csv not read	KmeansSolution2	fitting = 1, range = 0	not performed

Workflow of the sampling script:

Phase	Content
1	Set-up: parameters, frame reading, grid and domain codes, correlation matrix
2	LUCAS points to be retained (forced, take-all stratum)
3	Spatial parameters: legend.csv (TRUE) or fitting = 1, range = 0 (FALSE)
4	SamplingStrata frame, fitting/range per domain, precision constraints (CV = 5 %)
5	Initial solution by K-means, domain by domain (spatial or non-spatial)
6	Genetic algorithm (spatial or non-spatial)
6b	Only if TRUE: per-domain check of the spatial component on the GA strata
7	Sample selection, reports, figures, geopackages, strata polygons

Workflow of the kriging script:

Phase	Content
K1	Set-up: parameters, legend and survey reading, DEM at the survey points, zone of each point
K2	Variogram fitting (regression kriging, spherical model): national and per zone
K3	Fitting and range per soil unit; size of each soil unit and implied correlation
K4	Reliability assessment; legend.csv, variogram plots, text report

2. Requirements and installation

Software. R (4.1 or later; RStudio recommended). The path examples below are Windows-style.

R packages. Sampling script: sp, raster, ggplot2, dplyr, tidyr, SamplingStrata, clhs, sf, terra, readxl, parallel, doParallel. Kriging script: sf, terra, gstat. gstat is no longer needed by the sampling script.

Automatic installation. A block at the top of each script (before the library() calls) installs from CRAN what is missing (parallel is part of base R). SamplingStrata is handled separately: version 1.6 or later is required. If the package is missing or older, the sampling script installs it from GitHub with `pak::pak("barcaroli/SamplingStrata")` (this needs an internet connection).

To install manually, run:

```
install.packages(c("pak","sp","raster","ggplot2","dplyr","tidyr","clhs",
                   "sf","terra","readxl","gstat","doParallel"))
pak::pak("barcaroli/SamplingStrata")
```

Hardware. The genetic algorithm runs in parallel (parallel = TRUE), using the available cores. The example frame has about 10,000 cells; run time depends on the machine and on the GA settings (Section 5.2). With spatial_component = TRUE the spatial K-means and the spatial GA compute distances between all pairs of cells of a domain, which needs more memory and time than the non-spatial run.

With spatial_component = TRUE, Phase 6b uses two internal, non-exported functions of SamplingStrata (`SamplingStrata:::buildStrataDFSpatial` and `:::bethel_cpp`). Another version of the package may behave differently or fail there; record the version you use.

3. Input data

All inputs are read from one folder (Input/ in the repository). File names are hard-coded in the scripts; if you rename a file, edit the matching read.csv / st_read / read_excel / rast call.

File	Content	Required fields	Used by
frame_SML_1km_6_soil_units.csv	Sampling frame: one row per 1 km cell	POINT_ID, X_LAEA, Y_LAEA (EPSG:3035), ORIGINE, PH, PH_VAR, SOC, SOC_VAR, ARG, DEM, zona, soil_unit	sampling
Soil_observed_points_2022_Italy.xlsx	LUCAS points, sheet Points	POINT_ID, X_WGS84, Y_WGS84, SOIL (1 = to be forced)	sampling
legend.csv	Spatial parameters per soil unit (Section 3.1)	zona, soil_unit, fitting_PH, range_PH_km, fitting_SOC, range_SOC_km	written by kriging; read by sampling only if spatial_component = TRUE
Test_6_soil_units_Italy.gpkg	Legend: one polygon per soil unit	zona, soil_unit; CRS EPSG:3035	kriging (and the SML tool)
SharewithMS20260825-IT.csv	Point-level soil survey	X_LAEA, Y_LAEA, pH_H2O, SOC	kriging
DEM_3035_1000m.tif	1 km elevation raster, EPSG:3035	DEM covariate of the variogram trend	kriging

Notes on the frame:

- ORIGINE: cells with value LUCAS_2km are flagged as LUCAS-grid cells (LUCAS_GRID = 1).
- POINT_ID links LUCAS points to grid cells: LUCAS points must sit on grid nodes and share the ID.
- Domains are built as zona*soil_unit. The same key is built from the zona and soil_unit columns of legend.csv, so the names in the frame, in the legend .gpkg and in legend.csv must match exactly.
- Cells with non-finite PH, SOC, ARG, DEM or variances are dropped automatically (a message reports how many).
- Auxiliary variables with negative values are shifted to be non-negative (a warning is issued).

3.1 legend.csv

One row per soil unit. The sampling script reads only the columns below; any other column is ignored, except *reliability*, which is copied to the summary report and triggers a warning if it is not ADEQUATE. The file is checked when read: every soil unit of the frame must have exactly one row, fitting must be in (0, 1] and range must be >= 0.

Column	Meaning
zona, soil_unit	Zone and soil unit, as in the frame
fitting_<var>	Fitting (R2) of target <var> (PH, SOC), in (0, 1]. fitting = 0 is rejected by SamplingStrata.
range_<var>_km	Range of target <var>, in km (converted to metres in the script, like the grid coordinates)
reliability (optional)	ADEQUATE / USE WITH CAUTION / NOT RELIABLE, as written by the kriging script

Setting fitting = 1 and range = 0 for a soil unit makes that domain non-spatial while the others keep their spatial parameters. The kriging script also writes diagnostic columns (source_<var>, n_<var>, range_capped_<var>, mean_dist_km, rho_<var>, reliability_<var>), described in Section 6.4.

4. Preparing the run

4.1 Set the folders

The path variables near the top of **both** scripts must be changed: as delivered they point to a folder (SML_sampling_procedure_example) that differs from this repository.

```
inp_path <- "D:\\Google Drive\\ISPRA\\SML_sampling\\Input\\"
out_path <- "D:\\Google Drive\\ISPRA\\SML_sampling\\Output\\"
```

The other paths are derived from inp_path: `legend_csv_path` (both scripts) and `kriging_survey_path` (kriging script). If you change `legend_csv_path`, change it in both scripts. Keep the trailing separator on inp_path and out_path, because file names are appended with paste0() (forward slashes also work). The output folder must already exist: the scripts do not create it.

4.2 Run it

Open SML_sampling.Rproj, open a script and either press Source or run it section by section.

Non-spatial design (default): run the sampling script with `spatial_component <- FALSE`. The kriging script is not needed.

```
source("SML_sampling_example_with_6_Italy_soil_units.R", echo = TRUE)
```

Spatial design: (1) run the kriging script; (2) read `<base_nome>_kriging_report.txt` and check the reliability classes and the recommendation; (3) set `spatial_component <- TRUE` in the sampling script and run it.

```
source("Kriging_example_with_6_Italy_soil_units.R", echo = TRUE)
# edit spatial_component <- TRUE in the sampling script, then
source("SML_sampling_example_with_6_Italy_soil_units.R", echo = TRUE)
```

The genetic algorithm uses set.seed(1234), and the kriging script sets the same seed for the random points used to measure the soil units, so the runs are meant to be reproducible; change the seed to explore the variability of the solution.

5. Customising the scripts

5.1 Main parameters of the sampling script

The block "Parameters & settings" (about line 165) groups the choices users are expected to change.

Parameter	Default	Meaning / effect
min_point_per_domain	5	Domains with fewer cells are dropped (too small to stratify or cluster).

Parameter	Default	Meaning / effect
spatial_component	FALSE	TRUE: fitting/range per soil unit read from legend.csv and used in KmeansSolutionSpatial and optimStrata; Phase 6b performed. FALSE: KmeansSolution2, optimStrata with fitting = 1 and range = 0, no Phase 6b, legend.csv not read.
fitting_warn_threshold	0.9	With TRUE, a fitting below this value in legend.csv triggers a warning in the summary report.
vars_cont_y	PH, SOC	Target variables (frame column names; legend.csv columns fitting_<var>, range_<var>_km).
variance_cont_y	PH_VAR, SOC_VAR	Prediction variance of each target; same order as vars_cont_y.
vars_cont_x	ARG, DEM	Auxiliary variables used to build strata.
use_lucas_points	TRUE	TRUE: LUCAS points with SOIL == 1 are forced into the sample as a take-all stratum. FALSE: skip Phase 2.
use_lucas_grid	TRUE	TRUE: the LUCAS_GRID flag is carried in the frame, and sampled cells on the LUCAS grid are labelled sampled_lucas.
base_nome	Example_with_6_soil_units	Prefix for all output files.

5.2 Further settings inside the sampling code

Setting (where)	Default	Meaning
legend_csv_path (top)	inp_path/legend.csv	Location of legend.csv; must match the kriging script.
cv data frame (Phase 4)	0.05	Precision constraint (CV). Edit the rep(0.05, ndom) inside cv_cols to change it; one column per target (CV1, CV2) and one row per domain, so values can differ by domain and by variable.
max_k (Phase 5)	5	Upper bound of K-means clusters (initial strata) per domain.
iter, pops (Phase 6)	15, 10	GA generations and population size. Increase for a more thorough search (longer run time).
parallel (Phase 6)	TRUE	Set FALSE if the parallel back-end causes problems.
kappa	1	Passed to SamplingStrata spatial functions (correlation exp(-kappa d / range)); keep the same value in the kriging script.
res_m (final export)	1000	Cell size in metres for the strata polygons; must match the grid.

5.3 Parameters of the kriging script

Parameter	Default	Meaning
base_nome	Example_with_6_soil_units	Prefix of the variogram plots and of the report.
vars_cont_y, kriging_source_col	PH, SOC; c(PH="pH_H2O", SOC="SOC")	Target variables and the corresponding column of the point survey.
trend_covars	DEM for PH and SOC	Trend covariates added to x + y in the regression-kriging variogram (columns of the survey after the DEM extraction).
min_obs_district	30	Valid survey points a zone needs for its own variogram; otherwise the national model is used.
range_frac_max	0.8	Cap on the fitted range as a fraction of the largest empirical lag; a capped range is flagged.
fitting_low_threshold	0.5	Fitting below this: NOT RELIABLE.
fitting_warn_threshold	0.9	Fitting below this: USE WITH CAUTION.
rho_warn_threshold	0.5	Correlation at the mean within-unit distance above this: USE WITH CAUTION (large range).
kappa	1	Same value as in the sampling script.
n_pts_extent	500	Random points per soil unit used to compute its mean internal distance.

5.4 Adapting the scripts to another area or other variables

1. Build a new frame CSV with the same field names (or change the names in the code: X_LAEA, Y_LAEA, ORIGINE, zona, soil_unit, POINT_ID).
2. Update the legend .gpkg so that its zona and soil_unit values match the frame.
3. Provide the survey with the columns named in kriging_source_col and the coordinates X_LAEA, Y_LAEA, and re-run the kriging script so that legend.csv covers the new soil units.
4. To use other targets or auxiliaries, change vars_cont_y, variance_cont_y, vars_cont_x in the sampling script and vars_cont_y, kriging_source_col, trend_covars in the kriging script. Also update cor_vars (correlation matrix). Phase 6b and the reports adapt to the number of targets.
5. Update titles and text that mention PH, SOC, ARG, DEM, the two zones and "6 domains", if needed.
6. Change base_nome so earlier results are not overwritten.

The LUCAS step requires the point POINT_ID to appear in the frame. Points that do not match (outside the study area) or have missing values in the target, variance or auxiliary variables are discarded and counted in the console message.

6. Outputs

6.1 Sampling script

Files are written to out_path, with prefix base_nome. Start with the summary report.

File	Content
_summary.txt	Text report: spatial parameters used (or "not used"), optimisation (Phase 6), check of the spatial component (Phase 6b, if performed), warnings, final sample, expected CVs, sites per domain.
_units.csv	One row per domain: number of cells, validity flag, total sites, LUCAS-grid sites, forced LUCAS sites.
_allpoints.csv	All selected points (sampled + forced LUCAS): cell, domain, coordinates (EPSG:3035), stratum, source.
_sample.csv	SamplingStrata sample with sampling weights.
_figures.pdf	CV boxplot, map by soil unit, stacked barplot of points per domain, geographic distribution of points.
_correlation_matrix.pdf	Correlation matrix of PH, SOC, ARG, DEM.

File	Content
.gpkg	The 1 km grid with domain and sample counts per domain (GeoPackage).
.RData	Complete workspace, for later inspection.
strata_SML.gpkg	Optimised strata as polygons (fields soil_unit, STRATO, stratum_id); written without the prefix.

In the source column of _allpoints.csv: sampled = newly selected cell; sampled_lucas = selected cell lying on the LUCAS grid; lucas = existing LUCAS point forced into the sample.

6.2 Kriging script

File	Content
legend.csv (legend_csv_path, input folder)	Spatial parameters per soil unit (Section 3.1) and diagnostics: source_<var> (zone fit or national fallback), n_<var> (points of the model), range_capped_<var>, mean_dist_km, rho_<var>, reliability_<var>, reliability (worst over the variables).
_variograms.pdf	Empirical variograms and fitted models, national and per zone (zone plots only where the zone has its own fit).
_kriging_report.txt	Text report: settings; variogram fits (fitting, trend R2, range, capping, number of points, source); parameters per soil unit; reliability per soil unit and variable with the reasons; warnings; recommendation on spatial_component; notes.

6.3 How to read the sampling results

- **Sample size.** Section 2 of the summary gives the GA sample per domain (spatial or non-spatial, as set by the flag); with TRUE, section 3 compares, on the same strata, the spatial and the non-spatial allocation and shows which was kept. The final total in section 5 includes forced LUCAS points.
- **Expected CV.** The maximum over domains for PH and SOC should not exceed 0.05 (the boxplot in _figures.pdf has a red line at 0.05).
- **Warnings** (only with TRUE). Section 4 flags a low fitting in legend.csv, spatial allocations kept despite a low fitting, and soil units that the kriging script did not classify as ADEQUATE.
- **CV basis.** CVs are computed on the model-predicted values and variances in the frame, not on new field data.
- **Reference outputs.** The files in Output/ were produced by the previous version of the script (spatial fit computed inside it, GA always non-spatial). Re-run the scripts to obtain outputs consistent with this version.

6.4 How to read the kriging report

SamplingStrata (method = "spatial") computes the variance of a stratum h with n units from all pairs of its cells i, j at distance d_{ij}:

```
S2_h = [ sum_ij (z_i - z_j)^2 / fitting
       + sum_ij (s_i^2 + s_j^2 - 2 s_i s_j exp(-kappa d_ij / range)) ] / (2 n^2)
```

where z are the predicted values of the frame and s^2 their prediction variances. Two consequences guide the reliability rules:

- A **low fitting** inflates the first term: it states that the frame values explain little of the true variability.
- A **large range** compared with the distances inside a stratum makes exp(-kappa d / range) close to 1, so the prediction errors are treated as almost perfectly correlated and the second term nearly cancels. The sample size then decreases because of the range, independently of the fitting; if the range is not supported by local data, that reduction is not warranted. With range = 0 the exponential is 0 and the second term is the full prediction variance (non-spatial case).

For each soil unit the report gives the mean distance d between two random points of the unit and rho = exp(-kappa d / range). Strata are smaller than the soil unit, so the correlation inside a stratum is even higher than rho. Reliability classes:

Class	Condition
NOT RELIABLE	fitting < fitting_low_threshold (0.5)
USE WITH CAUTION	fitting < fitting_warn_threshold (0.9), or the zone uses the national model, or the range was capped at the data support, or rho > rho_warn_threshold (0.5)

Class	Condition
ADEQUATE	none of the above

The report recommends spatial_component = FALSE if any soil unit is NOT RELIABLE, and a comparison of the TRUE and FALSE runs if any is USE WITH CAUTION.

Example data. Both zones fall back to the national model (Emilia: no convergence of the zone fit with 100 / 61 valid points for PH / SOC; Toscana: 12 / 10 valid points). National fitting 0.41 (PH) and 0.49 (SOC), ranges 109 and 80 km, against mean distances of 11-72 km inside the soil units (rho 0.41-0.91). All six soil units are NOT RELIABLE; the sampling script is therefore delivered with spatial_component = FALSE.

Note on the fitting. The value written to legend.csv is the R2 of the variogram fit on the empirical semivariances. In the formula above the fitting divides the variance of the predicted values, i.e. it plays the role of the R2 of the model that produced the frame values; the R2 of the trend regression (x + y + DEM) is reported in the report for comparison (0.31 for PH, 0.27 for SOC in the example).

6.5 Using the strata in the SML tool

Copy strata_SML.gpkg to the Files/Strata/ folder of the SML tool and enter the settings in parameters_values_for_SML_tool.txt: soil unit file Test_6_soil_units_Italy.gpkg with ID field soil_unit; optional stratification from an existing polygon file (strata_SML.gpkg), column stratum_id, with areas not covered by the strata forming a separate stratum; point placement "simple random within strata".

7. Troubleshooting

Quick checklist: (1) paths updated in both scripts, (2) output folder exists, (3) packages installed, (4) spatial_component chosen, (5) if TRUE: kriging script run and its report read, (6) sampling script run, (7) summary and warnings read, (8) figures checked, (9) strata copied to the SML tool.

Symptom	Likely cause and remedy
cannot open file / cannot find path	Paths not updated, missing trailing separator, or the output folder does not exist (Section 4.1).
There is no package called ...	Install the package (Section 2).
spatial_component = TRUE but legend.csv was not found	Run the kriging script first, check legend_csv_path in both scripts, or set spatial_component <- FALSE.
legend.csv: missing column(s) / no row for the soil unit(s) / duplicated soil unit(s)	legend.csv does not match vars_cont_y or the frame: re-run the kriging script, or fix zona/soil_unit spelling by hand.
legend.csv: every fitting value must be in (0, 1]	A fitting of 0, negative, above 1 or missing. Use fitting = 1 and range = 0 to make a soil unit non-spatial.
stop(): fit_variogram_model() rejected the NATIONAL fit (kriging script)	The national variogram could not be fitted or has R2 <= 0. Inspect the survey data, the trend and the column in kriging_source_col; there is no further fallback.
Zone falls back to the national model (kriging script)	Fewer than min_obs_district valid points in the zone or a degenerate fit. Expected in the example; see the reliability report.
optimStrata refuses to run / non-finite values	Missing or infinite values in X, Y, variance or coordinates; the script removes them, but check the frame if many are dropped.
"nStrata is not feasible"	A domain is too small; increase min_point_per_domain (the script already caps strata at half the cells of the domain).
Error in the Phase 6b ::: lines	Different SamplingStrata version; the internal functions changed. Use the version the script was written with.
Memory errors / run too slow with TRUE	The spatial functions use all pairs of cells of a domain. Set parallel = FALSE, or reduce iter and pops for tests.
Slightly different totals between runs or machines	Independent Bethel calls differ by solver noise (reported in section 3 of the summary); keep the seed fixed to reproduce.